

		$= 9.088 \times 10^{-31} \text{ kg}$
29	B	Option A: Incorrect A large number of α-particles that were undeviated suggested that the atom consists of mostly empty space. Option B: Correct

Qn	Answer	Solution
		Since α-particles are positively charged, the small number of α-particles that were deviated by more than 90° showed that they had come very close to a small, massive positively charged nucleus at the centre of a gold atom. Option D: Incorrect The experiment had to be conducted inside an evacuated space because any deflection of the α-particles would only be due to collisions with the gold atoms and not due to collisions and scattering with the air particles. Option D: Incorrect The screen produced small flashes of light when the α-particles struck the screen is a direct result of the interaction between the screen and the α-particles, unrelated to nucleus of the gold atom.
30	A	Actual initial count rate of the radioactive sample